

Card Sort Report: Categories and Labels

Carley Andelin and Ashley Thornton

Brigham Young University

CS 356: Advanced Techniques in HCI

Dr. Mike Jones and Professor Bob Carroll

October 1, 2026

1. Who Was in the Study

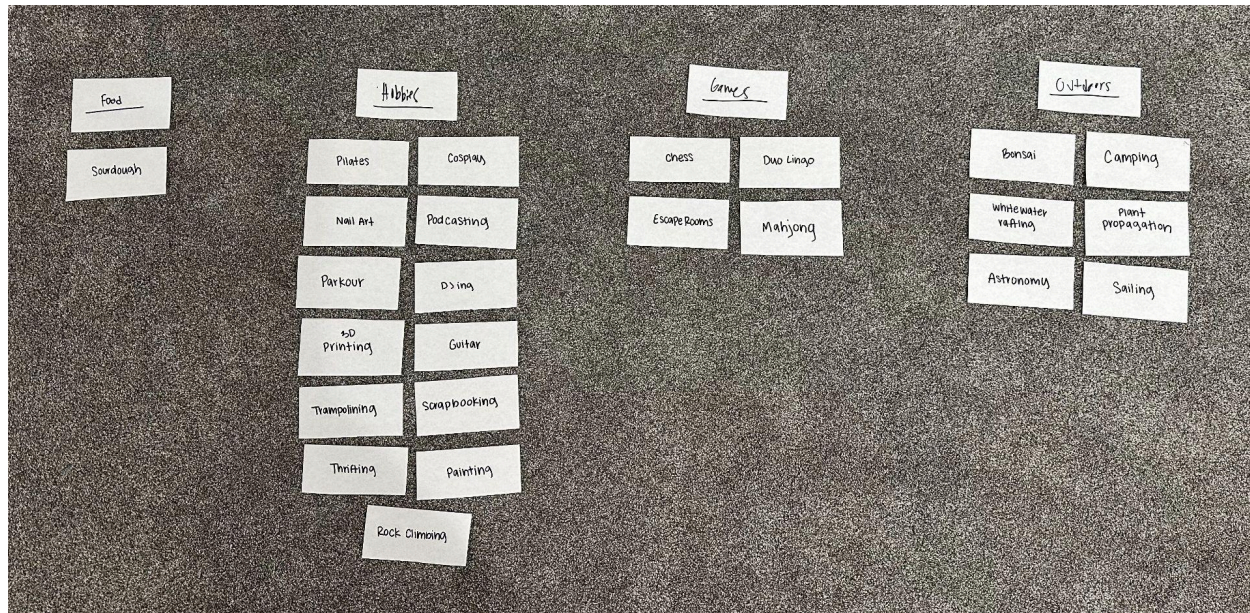
Sort	Participant	Interests	Selection Criteria
S1	Spencer Tagge (P01)	Outdoorsman, Hiking, Sports, Concerts, Gaming	Spencer is a curious person who wants to “expand his experiences” to live life to its fullest.
S2	Savannah Tagge & Carter Johnston (P02 + P03)	Savannah: Board games, Concerts Carter: Fitness, Gaming	They spend a lot of time together trying new activities (game nights, hikes, water sports).
S3	Kate Thornton (P04)	Music, Foodie, Arts and Crafts, Social Experiences	Kate manages a team of employees and is looking for team building/party activity ideas.
S4	Libby Tanner (P05)	Drawing, Graphic Design, Technology, Concerts	Libby is a busy student. When she does have free time, she doesn't know how to spend it.
S5	Paige Gordichuk (P06)	Music, trying new crafts, going out, problem solving	She likes to try new things to see if something enjoyable can fit into her busy schedule.
S6	Morgan Jenkins (P07)	Relaxing, trying new things, organization, baking	Loves the NYT game “connections”
S7	Meesa Andelin (P08)	Paddleboarding, games, baking, crafts	Seemingly has a new favorite hobby every month
S8	Savannah Sloan (P09)	Thrifting, doing nails, makeup, interior design	She is very trendy

Method

We ran an open card sort with 24 cards, each naming a hobby or activity (Table A1). Nine participants completed 8 sorts; Savannah Tagge and Carter Johnston sorted together as a pair. Each participant received the shuffled deck, grouped the cards however made sense to them, and named each group on a blank card. No categories were provided, and participants could merge, rename, or abandon groups at any point. We photographed each final sort, standardized similar labels into themes (Table A2), and tracked each card's placement across sorts (Table A3).

2. Pictures of the Resulting Sorts

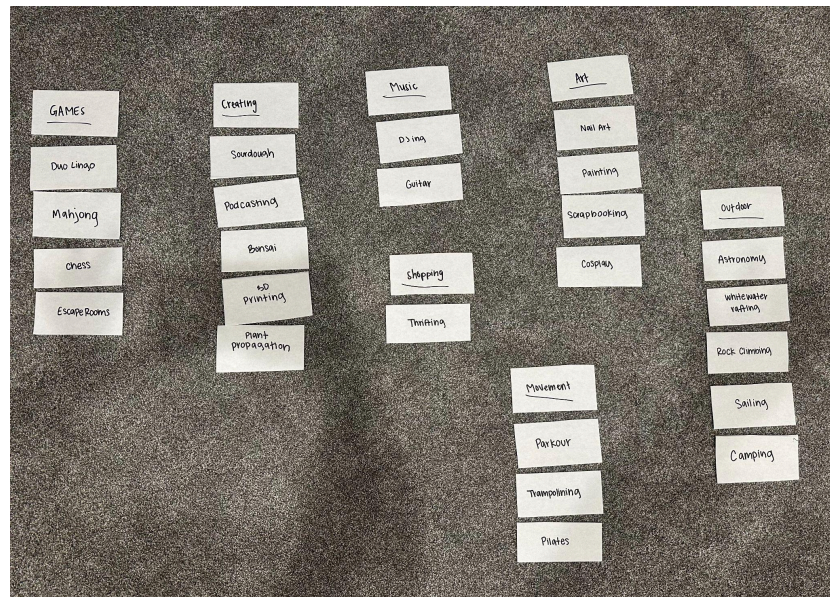
S1 P01 Spencer Tagge: Sorted by Location, Subject, Type of Activity, and a General Catch-All Category



"Food"	"Hobbies"	"Games"	"Outdoors"
1. Sourdough	1. Pilates 2. Cosplay 3. Nail Art 4. Podcasting 5. Parkour 6. DJing 7. 3D Printing 8. Guitar 9. Trampolining 10. Scrapbooking 11. Thrifting 12. Painting 13. Rock Climbing	1. Chess 2. Duo Lingo 3. Escape Room 4. Mahjong	1. Bonsai 2. Camping 3. Whitewater Rafting 4. Plant Propagation 5. Astronomy 6. Sailing

This sort had the fewest piles. It started with a catch-all label "activities" that was renamed to "hobbies". When changing that label Spencer said, "Actually I'm just going to call them hobbies". He pulled out a few noticable cards (chess, camping, sailing, etc.) from the catch-all to create "Games" and "Outdoors" groups. "Games" and "Outdoors" became two of the most frequently named labels throughout all 8 cardsorts.

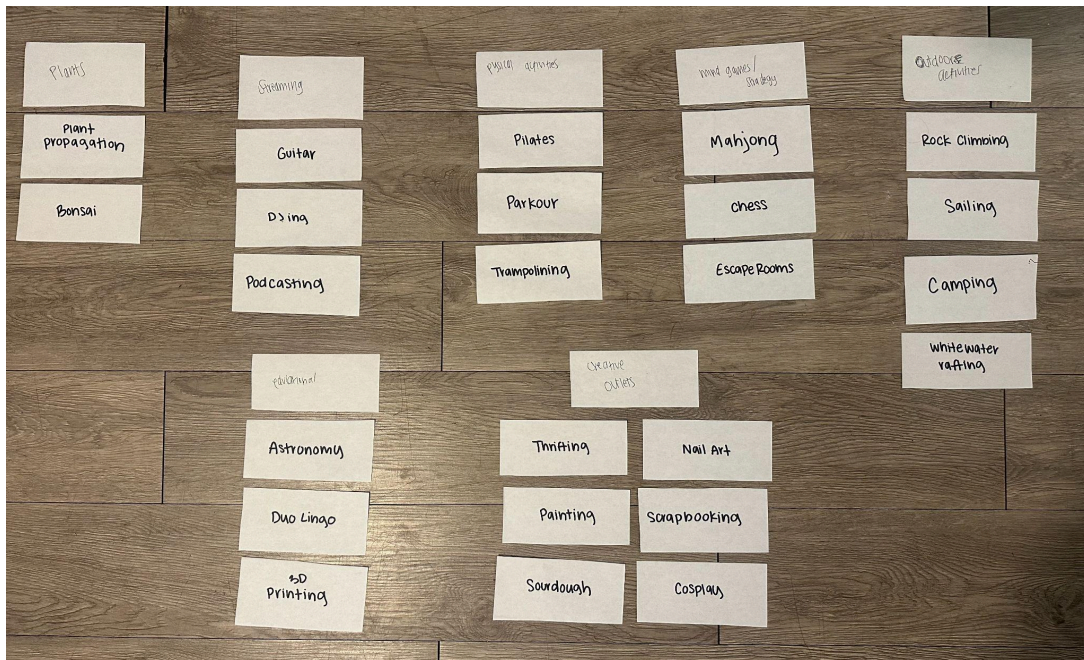
**S2 P02 & P03 Savannah Tagge and Carter Johnston: Sorted by Type of Activity,
Subject, Engagement, and Location**



"Games"	"Creating"	"Music"	"Shopping"
1. Duo Lingo 2. Mahjong 3. Chess 4. Escape Rooms	1. Sourdough 2. Podcasting 3. Bonsai 4. 3D Printing 5. Plant Propagation	1. DJing 2. Guitar	1. Thrifting
"Art"	"Movement"	"Outdoor"	
1. Nail Art 2. Painting 3. Scrapbooking 4. Cosplay	1. Parkour 2. Trampolining 3. Pilates	1. Astronomy 2. Whitewater Rafting 3. Rock Climbing 4. Sailing 5. Camping	

Instantly put together an "outdoors" category but then got stuck. Created a possible "creating" section as a solution. "Plant propagation is random but could kind of be creating". "Maybe an art section? For creative things that are specifically art?" Got stuck with trampolining and pilates which inspired a new "movement" section. They pulled "parkour" from "outdoors" to "movement".

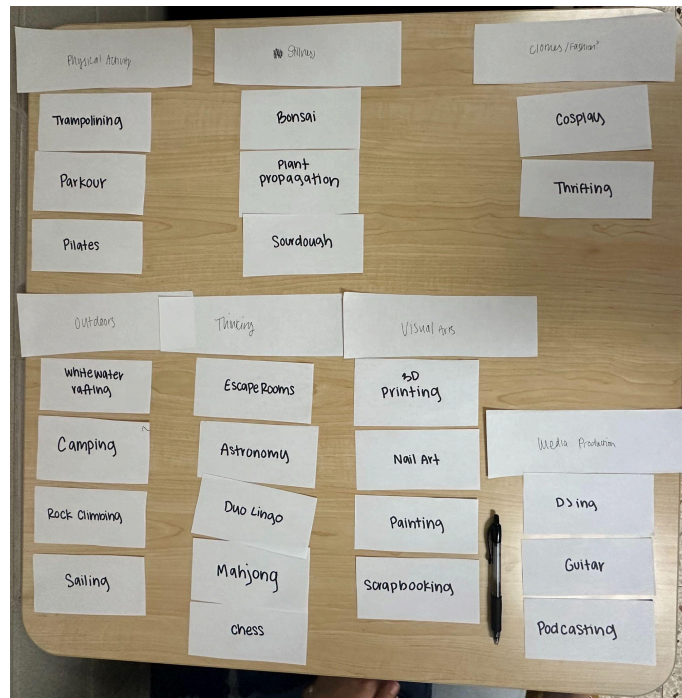
S3 P04 Kate Thornton: Sorted by Subject, Forms of Expression, Physicality vs. Mentality, and Location



"Plants"	"Streaming"	"Physical Activities"	"Mind games/strategy"
1. Plant Propagation 2. Bonsai	1. Guitar 2. DJing 3. Podcasting	1. Pilates 2. Parkour 3. Trampolining	1. Mahjong 2. Chess 3. Escape Room
"Outdoor Activities"	"Educational"		"Creative outlets"
1. Rock Climbing 2. Sailing 3. Camping 4. Whitewater Rafting	1. Astronomy 2. Duo Lingo 3. 3D Printing		1. Thrifting 2. Painting 3. Sourdough 4. Nail Art 5. Scrapbooking 6. Cosplay

Thought that rock climbing could be a part of the "physical/exercise" group but says "it could go in either physical or outdoors." Paused for a long time with astronomy duo lingo, guitar, dj, podcasting, cosplay, 3d printing. Then, pulled out a "streaming" group. Added sourdough and cosplay to arts, shifting it into more of a "creative outlet" group instead of just arts.

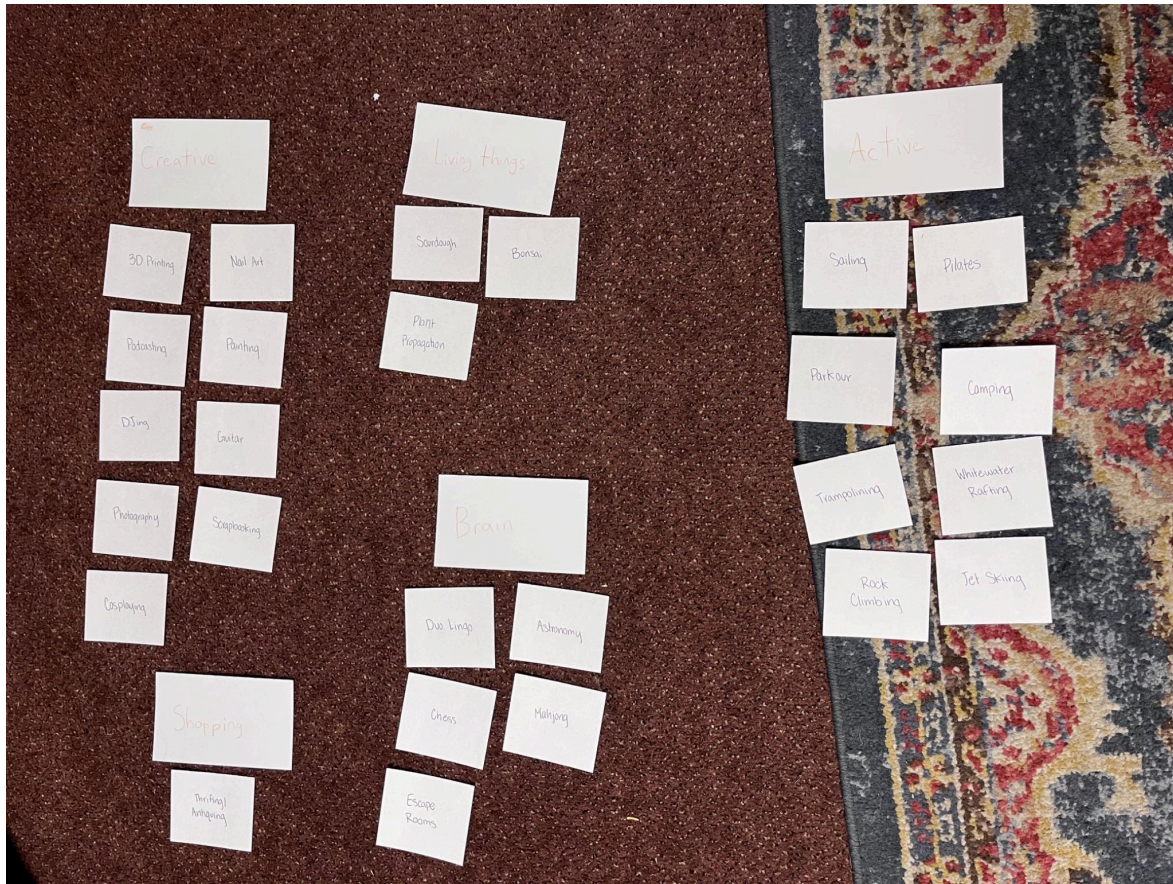
**S4 P05 Libby Tanner: Sorted by Physicality vs. Mentality, Internal Experience,
Location, and Artistry**



"Physical Activity"	"Stillness"	"Clothes/Fashion?"	"Outdoors"
1. Trampolining 2. Parkour 3. Pilates	1. Bonsai 2. Plant Propagation 3. Sourdough	1. Cosplay 2. Thrifting	1. Whitewater Rafting 2. Camping 3. Rock Climbing 4. Sailing
"Thinking"	"Visual Arts"	"Media Production"	
1. Escape Rooms 2. Astronomy 3. Duo Lingo 4. Mahjong 5. Chess	1. 3D Printing 2. Nail Art 3. Painting 4. Scrapbooking	1. DJing 2. Guitar 3. Podcasting	

Started with an outdoors group. Libby was grouping like-things in pairs and then got stuck after running out of pairs (djing and guitar, nail art and painting, bonsai and plant propagation, duo lingo and astronomy, trampolining and parkour). "Podcasting fits best with djing and guitar but I don't know why or what I would label it". "Sourdough goes with bonsai and plant propagation because they all require patience"

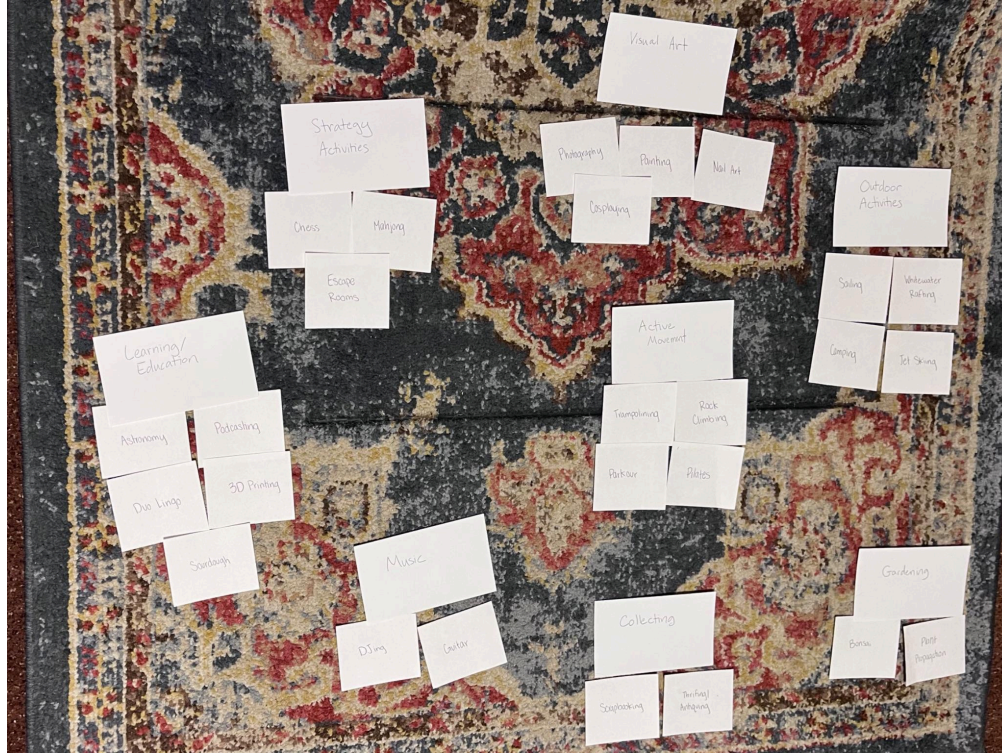
S5 P06 Paige Gordichuk: Sorted by Energy and Expression



"Creative"	"Living Things"	"Active"	"Shopping"	"Brain"
1. 3D Printing 2. Nail Art 3. Podcasting 4. Cosplaying 5. Painting 6. Guitar 7. Scrapbooking	1. Sourdough 2. Bonsai 3. Plant Propagation	1. Sailing 2. Pilates 3. Parkour 4. Trampolining 5. Camping 6. Whitewater Rafting 7. Rock Climbing	1. Thrifting/Antiquing	1. Duo Lingo 2. Astronomy 3. Chess 4. Mahjong 5. Escape Rooms

Immediately started sorting into two major groups, "Active" and "Creative". Once they were getting too big and ambiguous, the next idea to come was the "Brain" category. After that, it was clear that she wanted to place "Bonsai" and "Plant Propagation" together, and was unsure of where to put "Sourdough" and "Thrifting/Antiquing". Without wanting to make too many more groups, "Living Things" and "Shopping" came to be.

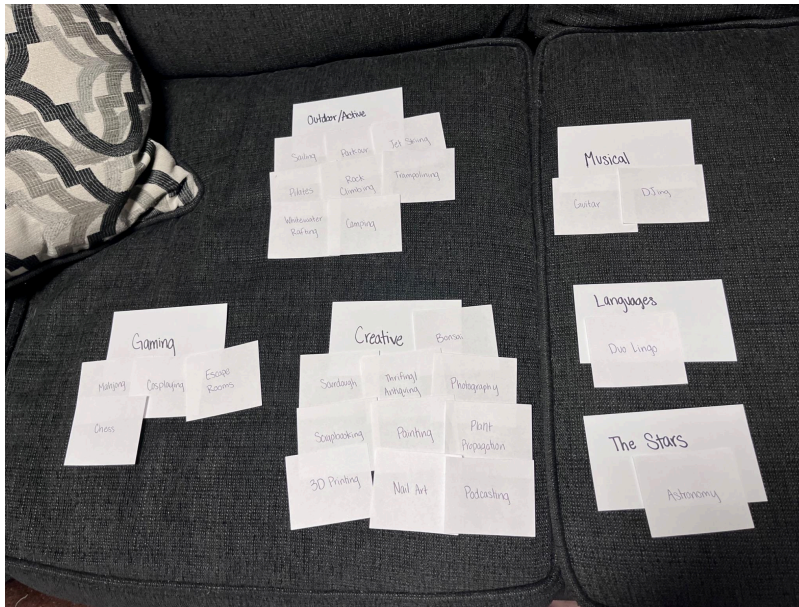
S6 P07 Morgan Jenkins: Sorted by Mental Activity, Physical Activity, and Expression



"Strategy Activities"	"Visual Art"	"Outdoor Activities"	"Active Movement"
1. Chess 2. Mahjong 3. Escape Rooms	1. Cosplaying 2. Painting 3. Nail Art	1. Sailing 2. Camping 3. Whitewater Rafting	1. Trampolining 2. Rock Climbing 3. Parkour 4. Pilates
"Learning/Education"	"Music"	"Collecting"	"Gardening"
1. Astronomy 2. Duo Lingo 3. Podcasting 4. 3D Printing 5. Sourdough	1. DJing 2. Guitar	1. Scrapbooking 2. Thrifting/Antiquing	1. Bonsai 2. Plant Propagation

Took some time to read through and look at all the cards first before trying to sort them into categories. Was willing to make more categories than P06 and did so with more precise labeling. After completing the card sort, she mentioned the only categories she didn't feel really good about were "Outdoor Activities" and "Active Movement" since the cards could fit into either category.

S7 P08 Meesa Andelin: Sorted by Activity and Expression

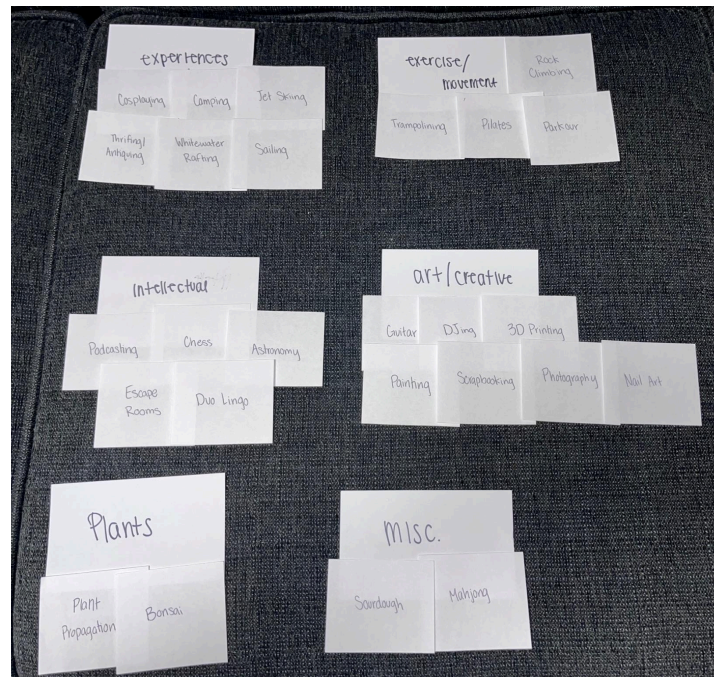


"Gaming"	"Outdoor/Active"	"Creative"
1. Mahjong 2. Chess 3. Cosplaying 4. Escape Rooms	1. Sailing 2. Parkour 3. Pilates 4. Rock Climbing 5. Trampoline 6. Whitewater Rafting 7. Camping	1. Sourdough 2. Thrifting/Antiquing 3. Scrapbooking 4. Painting 5. Plant Propagation 6. 3D Printing 7. Nail Art 8. Podcasting
"Musical"	"Languages"	"The Stars"
1. Guitar 2. DJing	1. Duo Lingo	1. Astronomy

Again, "Outdoor/Active" and "Creative" were the first groups to be categorized.

"Gaming" was next, followed by "Musical". Was stumped by where to put Duo Lingo and Astronomy, as she wanted them to fit into her already defined categories. After a few minutes of thought, the "Languages" and "The Stars" were made.

S8 P09 Savannah Sloan: Sorted by Expression, Engagement, and Other



"Experiences"	"Art/Creative"	"Intellectual"
1. Camping 2. Cosplaying 3. Thrifting/Antiquing 4. Whitewater Rafting 5. Sailing	1. Guitar 2. DJing 3. 3D Printing 4. Painting 5. Scrapbooking 6. Nail Art	1. Podcasting 2. Chess 3. Astronomy 4. Escape Rooms 5. Duo Lingo
"Exercise/Movement"	"Plants"	"Misc."
1. Rock Climbing 2. Trampolining 3. Pilates 4. Parkour	1. Plant Propagation 2. Bonsai	1. Sourdough 2. Mahjong

Once again, "Exercise/Movement" and "Art/Creative" were the first two categories to emerge. Many cards went into the "Exercise/Movement" group at first but then were removed and placed into the new "Experiences" category. After a few minutes of struggling with cards like Podcasting, Escape Rooms, Sourdough, Astronomy, etc., the "Intellectual" and "Misc." categories were created.

3. Analysis: Trends and Isolated Incidents

Participants created 4 to 8 groups each. Labels varied widely, but the piles were incredibly consistent.

Trends

1. **Several pairings appeared in all 8 sorts:** Camping/Sailing/Whitewater Rafting; Pilates/Parkour/Trampolining; Bonsai/Plant Propagation; DJing/Guitar; and Nail Art/Painting. Only the labels changed.
2. **Chess, Mahjong, and Escape Rooms** stayed together in 7 of 8 sorts (Mahjong went to "Misc." in Sort 8).
3. **Sorting started broad, then refined.** In Sorts 5, 7, and 8, an active/outdoor group and a creative group formed first. Leftover cards then prompted new groups, such as "Brain," "Languages," and "The Stars."
4. **Participants used different lenses:** location, body vs. mind, and type of expression, often mixing several in one sort.
5. **Astronomy and Duo Lingo** sat in Outdoors and Games in Sorts 1 and 2, but from Sort 3 on most participants placed them in a learning-type group ("Educational," "Thinking," "Brain," "Intellectual").

Isolated Incidents

1. **Sourdough** had no consistent home. It landed in a different group in nearly every sort, including "Food," "Creating," "Living Things," and "Misc."

2. **Thriftig** was placed in six different categories and was a single-card "Shopping" group in Sorts 2 and 5.
3. **Rock Climbing and Parkour** moved between outdoor and movement groups, and participants deliberated over both.
4. **One-off labels** included "Food," "Stillness," "Clothes/Fashion?," and "Misc."

4. Category and Label Decisions

We decided on 5 labels, two of which contain facets inside of themselves: growth and creative outlets. We prioritized labels that included the participants own words, however, “growth” came from patterns and themes we observed in the sorts rather than words spoken, or written, by the participants.

Label	Evidence	Why
Outdoors	6/8	Not only was this label named frequently, it was also always the first category the participants solidified and felt confident about. Location was a super strong scheme here that we decided needs to be included as it's own label.
Movement	7/8	There were a few splitters that were moved from “outdoors” to “movement” which told us that participants valued a separate “movement” group for specific terms (parkour, rock climbing).
Growth	8/8	Across all sorts the participants consistently grouped learning, and developing skills or things. This broader theme appeared through intellectual growth and plant growth. We decided to use those as two facets to the broader “growth” scheme.
Learning	6/8	Had different labels (intellectual, languages, the stars, education, brain, thinking) but were

		always referring to the same pile
• Cultivating	4/8	A facet under “Growth”. This label came from 2 “plants” and 1 “gardening” label that featured the “plant propagation” and “bonsai” in 8/8 sorts. “Gardening” may seem like a more obvious label choice, however “gardening” is a hobby included in our IA that will be included later so we decided that “Cultivating” relayed the same message and worked with that constraint.
Games	7/8	The results showed a very strong, consistent grouping of the terms “Chess, Mahjong, Escape Rooms”. Even when labels slightly differed (games, strategy, brain, thinking), the terms were almost always paired together.
Creative Outlets	8/8	In each of the card sorts, every participant began with some version of a “creative” category. 5/8 of the card sorts kept a label containing the word “creative”. The other $\frac{3}{8}$ broke “creative” into smaller facets (Music, art, visual art, clothes/fashion, media production)
• Art	3/8	“Maybe I should create an ‘art’ section for artsy creative things specifically.” This will be a facet under ‘Creative Outlets’.
• Media	5/8	“Djing” and “Guitar” were grouped together in 8/8 sorts and were commonly pulled from the “creative” category to form their own “music” or “media production” categories. Thus the basis for “media” as a facet to “creative outlets”.
• Collecting	1/8	“Collecting” will exist as a facet under “Creative Outlets” because of the splitter “thrifting/antiquing”. Savannah and Carter decided “thrifting needs to be its own section” in card sort #2 and that consensus rang true for the following cart sorts too. Across 8 studies it was sorted into 6 different categories with “creative” and “shopping” being it’s two repeat placements. (creativex2, experiences, collecting, shoppingx2, fashion). We decided to prioritize the “collecting” label over “shopping” in anticipation of other terms not included in the

		study that will fall under that category in future prototypes (watch collecting, baseball cards, pokemon cards, etc.).
--	--	--

One more decision that isn't a category: monitoring "outdoors" and "movement".

Another decision we made is to monitor the relationship between "outdoors" and "movement" during tree tests and future prototype tests. We observed an overlap between the two categories because of splitters like "rock climbing" and "parkour." While our data showed that participants valued separate "outdoor" and "physical" groups, we also observed a lot of deliberation from participants as they sorted activities into these two categories. This is evidence that the categories could potentially be consolidated in the future. A comment from Kate in Sort #2 inspired this decision: "Rock climbing could go in either physical or outdoors." If testing goes well, the two separate categories will remain intact. However, if participants struggle with the distinction, "outdoors" could potentially become a filter for the "movement" label.

5. What We'll Ignore for Now

Write a list about what we're going to ignore and why. How long are we going to ignore it? Just for now, or forever?

Appendix

Table A1

The Deck

Card
Plant Propagation
Mahjong
Camping
3D Printing
Scrapbooking
Duo Lingo
Parkour
Sailing
Podcasting
Cosplaying
Sourdough
Astronomy
DJing
Rock Climbing
Nail Art
Escape Rooms
Bonsai
Trampolining
Whitewater Rafting
Pilates
Painting
Guitar
Thrifting
Chess

Note. This table shows the terms used in all 8 card sorts.

Table A2

Standardizing Reoccurring Themes

Recurring theme	Sort 1	Sort 2	Sort 3	Sort 4	Sort 5	Sort 6	Sort 7	Sort 8
Games / mind strategy	Games	Games	Mind Games	Thinking	Brain	Strategy Activities	Gaming	Intellectual
Outdoors / adventure	Outdoors	Outdoors	Adventure activities	Outdoors	Active	Outdoor Activities	Outdoor/Active	Experiences
Movement / physical	Hobbies	Movement	Physical activities	Physical Activity	Active	Active Movement	Outdoor/Active	Exercise/Movement
Creative / visual arts	Hobbies	Art / Creating	Creative outlets	Visual Arts	Creative	Visual Art	Creative	Art/Creative
Music / media	Hobbies	Music	Streaming	Media Production	Creative	Music	Musical	Art/Creative
Plants / patience	Outdoors	Creating	Plants	Patience-related	Living Things	Gardening	Creative	Plants
Learning / education	Not a distinct group in Sorts 1-4 (Duo Lingo sits with Games, Astronomy with Outdoors or Thinking)				Brain	Learning/Education	Languages / The Stars	Intellectual

Note. This table groups together similar themes and labels into more standard groups.

Table A3

Each Term and Which Groups They Were Placed In

Card	Sort 1	Sort 2	Sort 3	Sort 4	Sort 5	Sort 6	Sort 7	Sort 8
Plant Propagation	Outdoors	Creating	Plants	Patience-related	Living Things	Gardening	Creative	Plants
Mahjong	Games	Games	Mind Games	Thinking	Brain	Strategy Activities	Gaming	Misc
Camping	Outdoors	Outdoors	Adventure activities	Outdoors	Active	Outdoor Activities	Outdoor/Active	Experiences
3D Printing	Hobbies	Creating	Hobbies (temporary)	Visual Arts	Creative	Learning/Education	Creative	Art/Creative
Scrapbooking	Hobbies	Art	Creative outlets	Visual Arts	Creative	Collecting	Creative	Art/Creative
Duo Lingo	Games	Games	Hobbies (temporary)	Thinking	Brain	Learning/Education	Languages	Intellectual
Parkour	Hobbies	Movement	Physical activities	Physical Activity	Active	Active Movement	Outdoor/Active	Exercise/Movement
Sailing	Outdoors	Outdoors	Adventure activities	Outdoors	Active	Outdoor Activities	Outdoor/Active	Experiences
Podcasting	Hobbies	Creating	Streaming	Media Production	Creative	Learning/Education	Creative	Intellectual
Cosplaying	Hobbies	Art	Creative outlets	Clothes/Fashion?	Creative	Visual Art	Gaming	Experiences
Sourdough	Food	Creating	Creative outlets	Patience-related	Living Things	Learning/Education	Creative	Misc
Astronomy	Outdoors	Outdoors	Hobbies (temporary)	Thinking	Brain	Learning/Education	The Stars	Intellectual
DJing	Hobbies	Music	Streaming	Media Production	Creative	Music	Musical	Art/Creative
Rock Climbing	Hobbies	Outdoors	Adventure activities	Outdoors	Active	Active Movement	Outdoor/Active	Exercise/Movement
Nail Art	Hobbies	Art	Creative outlets	Visual Arts	Creative	Visual Art	Creative	Art/Creative
Escape Rooms	Games	Games	Mind Games	Thinking	Brain	Strategy Activities	Gaming	Intellectual
Bonsai	Outdoors	Creating	Plants	Patience-related	Living Things	Gardening	Creative	Plants
Trampolineing	Hobbies	Movement	Physical activities	Physical Activity	Active	Active Movement	Outdoor/Active	Exercise/Movement
Whitewater Rafting	Outdoors	Outdoors	Adventure activities	Outdoors	Active	Outdoor Activities	Outdoor/Active	Experiences
Pilates	Hobbies	Movement	Physical activities	Physical Activity	Active	Active Movement	Outdoor/Active	Exercise/Movement
Painting	Hobbies	Art	Creative outlets	Visual Arts	Creative	Visual Art	Creative	Art/Creative
Guitar	Hobbies	Music	Streaming	Media Production	Creative	Music	Musical	Art/Creative
Thrifting	Hobbies	Shopping	Creative outlets	Clothes/Fashion?	Shopping	Collecting	Creative	Experiences
Chess	Games	Games	Mind Games	Thinking	Brain	Strategy Activities	Gaming	Intellectual

Note. This table shows every term in the deck and which category they were placed in by participants in each sort.